# POTHURI ASRITH

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## PROFESSIONAL SUMMARY

- Computational Structural Engineer with 4.6+ years of experience combining structural engineering expertise with Python-based automation, AI, and digital engineering technologies to optimize design processes, improve accuracy, and accelerate project delivery across complex building developments
- Skilled in Python, AI, Machine Learning (ML), MCP, Grasshopper, ETABS, Revit, AutoCAD, RCC Structural Design, and BIM technologies, with hands-on experience developing automation solutions that reduce manual effort, improve accuracy, and accelerate engineering deliverables
- Strong foundation in RCC structural analysis, design optimization, Eurocode-based design principles, computational engineering, and international design standards, supported by advanced academic qualifications from a Washington Accord-recognized institution and practical experience across high-rise and large-scale building developments
- Recognized for analytical thinking, attention to detail, collaboration, and initiative, with a proven ability to work across multidisciplinary teams, solve engineering challenges, and deliver reliable, technology-driven solutions that create measurable value for projects and organizations

## SKILL SET

- **Computational Structural Engineering:** Structural Design, Structural Analysis, RCC Design, High-Rise Building Design, FEA Modeling, Seismic & Wind Analysis, ACI Code, Eurocode
- **Programming & Automation:** Python, Engineering Automation, ETABS API, Revit API, Workflow Automation, MCP, Grasshopper
- **BIM & Digital Engineering:** ETABS, Revit, AutoCAD, BIM Coordination, Computational Design
- **AI & Emerging Technologies:** Artificial Intelligence (AI), Machine Learning (ML), Parametric Modeling

## SOFT SKILLS

- Project Coordination, Cross-Functional Collaboration, Stakeholder Management, Process Improvement, Analytical Thinking, Problem Solving, Attention to Detail, Communication, Ownership

## EDUCATION

- Master of Technology in Computer-Aided Structural Engineering – Mahindra University, Hyderabad, India | Sep 2021 – Sep 2023
- Bachelor of Technology in Civil Engineering (Honours) – Manipal International University, Malaysia | June 2021 (Academic curriculum focused on Eurocode-based structural design and analysis methodologies)

## WORK EXPERIENCE

**Computational Design Engineer, Dar Al Handasah, India | Aug 2025 – Present**

- Develop Python-based automation frameworks that streamlined structural modeling, Revit drafting, and QA/QC processes across projects
- Engineer beam and column automation tools that transformed ETABS analysis models into production-ready CAD drawings, eliminating manual drafting effort
- Integrate ETABS and CAD data streams to enable real-time member verification and accelerate the detection of modeling discrepancies
- Lead the development, testing, and deployment of Revit add-ins and ETABS automation solutions supporting RCC building design, ACI Code and Eurocode-compliant workflows, improving design accuracy and workflow efficiency

**Tech Stack:** Python, ETABS API, AutoCAD Automation, Data Validation Frameworks

**Structural Design Engineer II, Walter P Moore, India | Oct 2024 – July 2025**

- Delivered RCC structural design and analysis for high-rise and complex building projects using advanced FEA methodologies and international design standards
- Performed load calculations, member sizing, and structural system design for residential and commercial developments

- Maintained data APIs linking ETABS, Revit, and documentation platforms, ensuring seamless information flow across design teams
- Collaborated with architects, MEP consultants, and contractors to successfully execute projects valued at over $25M while meeting quality and schedule expectations

**Tech Stack:** ETABS, Revit, AutoCAD, Python, API Development

**Structural Designer, Design Werkz, India | July 2022 – June 2023 (Internship) | June 2023 – Oct 2024**

- Designed 30+ storey RCC and steel high-rise buildings incorporating shear walls, outriggers, and transfer structures in compliance with applicable design codes and large-scale development requirements
- Conducted seismic and wind analyses in accordance with IS 875 and IS 1893 standards, ensuring code compliance and structural reliability
- Optimized structural schemes through value engineering initiatives that reduced material costs by 15-20% without compromising performance
- Prepared detailed structural models, engineering calculations, and construction documentation to support efficient project delivery

**Tech Stack:** ETABS, Revit, AutoCAD, Manual Calculations, Structural Analysis

## INDEPENDENT COMPUTATIONAL PROJECTS

*Engineering Automation & AI Projects*

**AI-Powered Grasshopper Assistant for Rhino | Python, LLMs, MCP**

- Developed an AI assistant integrated with Rhino and Grasshopper to generate, explain, and modify parametric definitions using natural-language prompts.
- Built a custom Rhino plugin with MCP integration, supporting local LLMs (Ollama) and cloud-based AI models through a desktop interface.
- Prioritised native Grasshopper components and firm-specific libraries, using scripted components only when necessary to improve workflow efficiency.
- Benchmarked the solution against commercial Grasshopper AI tools across 100+ test cases, reducing script setup and modification time, and showcased it on LinkedIn and engineering platforms.

**Excel-to-AutoCAD Beam Schedule Automation Tool | Python**

- Developed an end-to-end Python solution to automatically convert Excel-based beam schedules into parametric AutoCAD drawings
- Automated schedule parsing, coordinate generation, layer management, and block placement workflows
- Eliminated 6-8 hours of manual drafting effort per project while reducing copy-paste errors
- Demonstrated the solution on LinkedIn and professional platforms, engaging 500+ engineering professionals

**AI-Based Concrete Strength Prediction Model | Python, Scikit-Learn, TensorFlow**

- Built a machine learning model using 900+ real-world concrete test samples to predict compressive strength
- Incorporated more than 10 material and environmental parameters, including water-cement ratio, curing duration, temperature, humidity, and admixture content
- Achieved 92% prediction accuracy, supporting more informed concrete mix design and quality control decisions
- Showcased the model on LinkedIn and professional engineering platforms

**ETABS Property Validator | Python, ETABS API**

- Developed an automated validation tool to verify ETABS section property naming conventions against actual geometric dimensions
- Identified naming inconsistencies, dimensional mismatches, and potential modeling errors before analysis
- Processed over 1,000 section properties in less than 30 seconds, reducing QA/QC review time by 80%
- Helped minimize rework, prevent analysis inaccuracies, and improve model reliability on complex projects
- Demonstrated the solution on LinkedIn and professional engineering platforms